Math 163 — Exam 1B — Answer Key

J-number 00499317

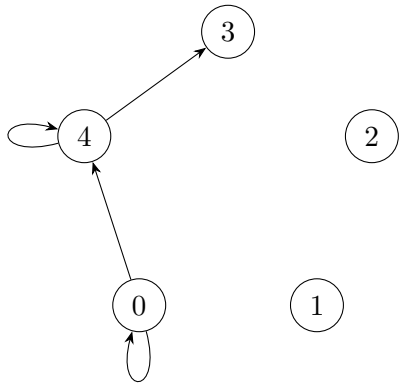
Setup

$J_1..J_8 = 0,0,4,9,9,3,1,7 \qquad r_1..r_8 = 0,0,4,4,4,3,1,2$
$\mathcal{U} = \{0,..,9\}, \quad J = \{0,1,3,4,7,9\}, \quad E = \{0,1,4,9\}, \quad s = 1, \quad b = 10, \quad A = \{0,1,10\}, \quad B = \{0,1,4,9\}$

$$R = \{(0,0)\} \cup \{(r_2,r_3),(r_3,r_4),(r_4,r_5),(r_5,r_6)\} = \{(0,0),\ (0,4),\ (4,3),\ (4,4)\} \ \subseteq \{0,1,2,3,4\}^2$$

1(abc)

$$M(R) = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \qquad M(R^{op}) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$



$DG(R)$: self-loops at $0, 4$; edge $0 \to 4$; edge $4 \to 3$; $1, 2$ isolated.

(iv) $R \circ R^{op} = \{(0,0),(0,4),(3,3),(3,4),(4,0),(4,3),(4,4)\}, \quad R^{op} \circ R = \{(0,0),(0,4),(4,0),(4,4)\}$. Both matrix identities verify: $(M(R \circ R^{op}))(0,1) = ((M(R^{op}))(M(R)))(0,1)$ and $(M(R^{op} \circ R))(0,1) = ((M(R))(M(R^{op})))(0,1)$ — confirmed by direct computation (the raw integer products have a stray 2 at one entry each, which the $(0,1)$-threshold removes to match).

(v) $FlxClsr(R) = R \cup \Delta_{0..4} = \{(0,0),(0,4),(1,1),(2,2),(3,3),(4,3),(4,4)\}$ — true by definition.
(vi) $SymClsr(R) = R \cup R^{op} = \{(0,0),(0,4),(3,4),(4,0),(4,3),(4,4)\}$ — true by definition.
(vii) $TrnsClsr(R) = \bigcup_{k=1}^{5} R^{\circ k} = \{(0,0),(0,3),(0,4),(4,3),(4,4)\}$ (stabilizes at $k = 2$; pigeonhole on $\nu(\mathcal{U}) = 5$ justifies the bound).
(viii)–(ix) $EqvClsr(R) = \{0,3,4\}^2 \cup \{(1,1),(2,2)\}$, matching $\bigcup_{k=0}^{5} (R \cup R^{op})^{\circ k}$. Quotient: $\dfrac{\{0,1,2,3,4\}}{EqvClsr(R)} = \big\{\{0,3,4\},\{1\},\{2\}\big\}$.

2a

$A = \{0, 1, 10\}$ ($\nu = 3$), $B = \{0, 1, 4, 9\}$ ($\nu = 4$).
(i)–(iv) All four are the same proposition $\exists f \in Fnc(A, B)(Inj(f) \wedge Srj(f))$, renamed. $\boxed{\text{N}}$ for all: since $\nu(A) = 3 < 4 = \nu(B)$, $\nu(Im(f)) \leq \nu(A) = 3$ for every f, so $Srj(f)$ (which needs $\nu(Im(f)) = 4$) is impossible.
(v)–(vi) Taking the natural order-preserving $\delta : A \to B$, $\delta(0)=0, \delta(1)=1, \delta(10)=1$: $Im(\delta) = \{0, 1\}$; maximal bijective restrictions are $\delta|_{\{0,1\}}$ and $\delta|_{\{0,10\}}$ — count = 2. *(The exam doesn't pin down δ explicitly — substitute your notes' convention if different.)*

2b

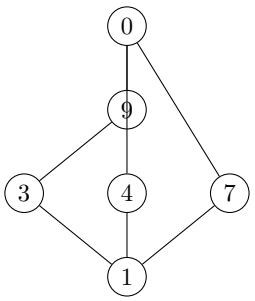
$R := \{(j, k) \in J \times J \mid 2 \mid |j-k|\}$ (same parity). $\boxed{\text{Y}}$ $Eqv(R)$: reflexive/symmetric/transitive all hold. Classes: $\{0, 4\}$, $\{1, 3, 7, 9\}$.
$S := \{(e, f) \in E \times E \mid (e \geq f) \vee (e \mid f)\}$. $\boxed{\text{N}}$ $PO(S)$: $(1, 4), (4, 1) \in S$ with $1 \neq 4$ (since $1 \mid 4$ and $4 \geq 1$) — antisymmetry fails.

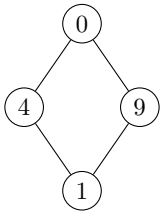
2c

Given $PO(T, \leq)$, claim $\forall \varphi \in Fnc(S, T)$ $PO(R(\varphi))$ where $R(\varphi) := \{(u, v) \mid \varphi(u) = \varphi(v)\}$. $\boxed{\text{N}}$: $R(\varphi)$ is always reflexive+symmetric (so always Eqv), but a constant φ (with $\nu(S) \geq 2$, $u \neq v$) gives $(u, v), (v, u) \in R(\varphi)$ — antisymmetry fails, so $PO(R(\varphi))$ fails for that φ.

3a

$P := \{(j, k) \in J \times J \mid j \mid k\}$. $\boxed{\text{Y}}$ $PO(P)$. Hasse: 1 covered by $3, 4, 7$; 9 covers 3; 0 covers $4, 7, 9$.
$Q := \{(e, f) \in E \times E \mid f \mid e\}$. $\boxed{\text{Y}}$ $PO(Q)$. Hasse: 1 covered by $4, 9$; 0 covers $4, 9$.





3b

Given S and equivalence relations E, F: $\boxed{\text{N}}$ for both (i) $Eqv(E \backslash F)$ and (ii) $Eqv((E \backslash F) \cup (F \backslash E))$. F (resp. $E \cap F$) always contains Δ_S, so removing it (resp. taking symmetric difference) always strips out every diagonal pair — reflexivity fails for every choice of S, E, F (nonempty S).

3c

$B = \{0, 1, 4, 9\}$, $\nu(B) = 4$.

$$\nu(PrtOrdrs(B)) = 219 \qquad \nu(TtlOrdrs(B)) = 24 = 4! \qquad \nu(EqvRlns(B)) = \nu(Prtns(B)) = 15$$

(the 219 figure is verified by exhaustive brute-force check, and follows the known sequence $1, 1, 3, 19, 219, 4231, \ldots$ for partial orders on $0, 1, 2, 3, 4, 5$-element sets — too many to list by hand, unlike the other three).
$TtlOrdrs(B)$: all 24 permutations of $\{0, 1, 4, 9\}$ read as chains, e.g. $0<1<4<9$, $0<1<9<4$, ...

$Prtns(B)$ (all 15; $EqvRlns(B)$ is the same 15, read as relations):

$$\{\{0,1,4,9\}\};\quad \{\{0\},\{1,4,9\}\},\{\{0,1\},\{4,9\}\},\{\{0,4,9\},\{1\}\};$$

$$\{\{0\},\{1\},\{4,9\}\},\{\{0,1,4\},\{9\}\},\{\{0,9\},\{1,4\}\};\ \{\{0\},\{1,4\},\{9\}\},\{\{0,4\},\{1,9\}\},\{\{0,1,9\},\{4\}\};$$

$$\{\{0\},\{1,9\},\{4\}\},\{\{0,1\},\{4\},\{9\}\},\{\{0,4\},\{1\},\{9\}\};\quad \{\{0,9\},\{1\},\{4\}\},\{\{0\},\{1\},\{4\},\{9\}\}$$